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Blue economy skills survey
Presentation of results and key findings

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1. Introduction

The ocean, seas and coastal areas provide a wealth of opportunities for economic growth, improved livelihoods, resilience, innovation, sustainability transition, and employment creation. Yet today the blue economy is challenged by an ageing workforce and a lack of workers for many related economic activities, including in emerging sectors such as Marine Renewable Energy and Blue Biotechnology.

A strategic approach to solving blue skills shortages in the EU is indispensable. With this in mind, the European Commission - Directorate-General for Maritime Affairs and Fisheries has decided to launch a one-year study to analyse the blue economy industry needs, existing academic/training programmes, and consequent gaps, so as to help inform a better policy response at EU level and provide very valuable information to Member States.

This survey aimed to gather valuable insights from stakeholders in the blue economy and collect comprehensive, reliable, and up-to date data to enhance understanding of the current skills landscape, future skills requirements, existing gaps, and potential solutions within the European Union's blue economy.

1.1 Objectives of the survey

The survey was launched on 15 September 2024 and remained open for a duration of eight weeks. Specifically, the survey aimed to:

- Establish a shared understanding of key concepts such as "blue skills" and the "blue economy".
- Map the current skills profiles of the blue economy workforce, including workforce composition, the most prevalent occupations, and in-demand skillsets across various maritime industries and EU regions. Additionally, the survey sought to anticipate future skills needs and occupational shifts over the next 5-10 years.
- Identify existing education and training programmes related to blue skills acquisition, encompassing formal degree programmes, vocational training, upskilling courses, and other capacity-building initiatives.
- Identify the most significant skills gaps, mismatches, and deficiencies currently affecting the EU's blue economy, along with the underlying drivers, barriers, and consequences of these deficits.

1.2 Methodology of the survey

The survey was disseminated through a combination of online and direct channels, including email invitations to key stakeholders, posts on social media platforms, and distribution via institutional networks connected to the blue economy. The response collection remained open for eight weeks, allowing sufficient time to maximize participation from professionals, academics, and organisations operating in maritime and coastal sectors. A **total of 233 respondents** participated, representing a wide range of roles, sectors, and geographical

regions. The survey results were further enriched with in-depth notes and additional data analysis to provide a holistic understanding of the blue economy's skills landscape.

The survey was structured into six distinct sections, each designed to gather comprehensive insights from respondents on various aspects of blue skills development:

- Identification: this section aimed to collect demographic information about respondents. Rather than identifying individuals, the goal was to categorize the population consistently. Given that the survey was not going to be statistically representative, it was crucial to control for variables such as age group, stakeholder type, blue economy sector, years of expertise, etc., to mitigate self-selection bias. Additionally, respondents could complete the survey anonymously, without providing any identifiable information, or choose to leave their contact details if they wished to be further involved in the project (e.g., to clarify points or participate in focus groups).

- Current skills profile and future needs: this section sought to better understand the skills currently required by workers in the blue economy and those anticipated to be needed in the future. Respondents rated the importance of various skills across three time horizons: short-term (0–2 years), medium-term (3–5 years), and long-term (6–10 years). To facilitate participation and ease of analysis, a closed list of skills was provided. Respondents also had the opportunity to flag additional skills not included in the list through an open-ended question.

- Education, training and re-skilling/upskilling programmes: this section evaluated awareness, participation, and the effectiveness of training programmes. Key aspects included cost-effectiveness, accessibility, practical learning opportunities, and alignment with industry needs.

- Barriers and solutions: this section explored challenges in skills development, such as funding limitations, lack of career pathways, and insufficient collaboration. Respondents were also encouraged to propose actionable solutions to address these challenges.

- Definition of "blue skills": this section presented a draft definition of "blue skills" and invited feedback and suggestions from respondents. Additionally, a space was provided for participants to share further comments or observations on blue skills development.

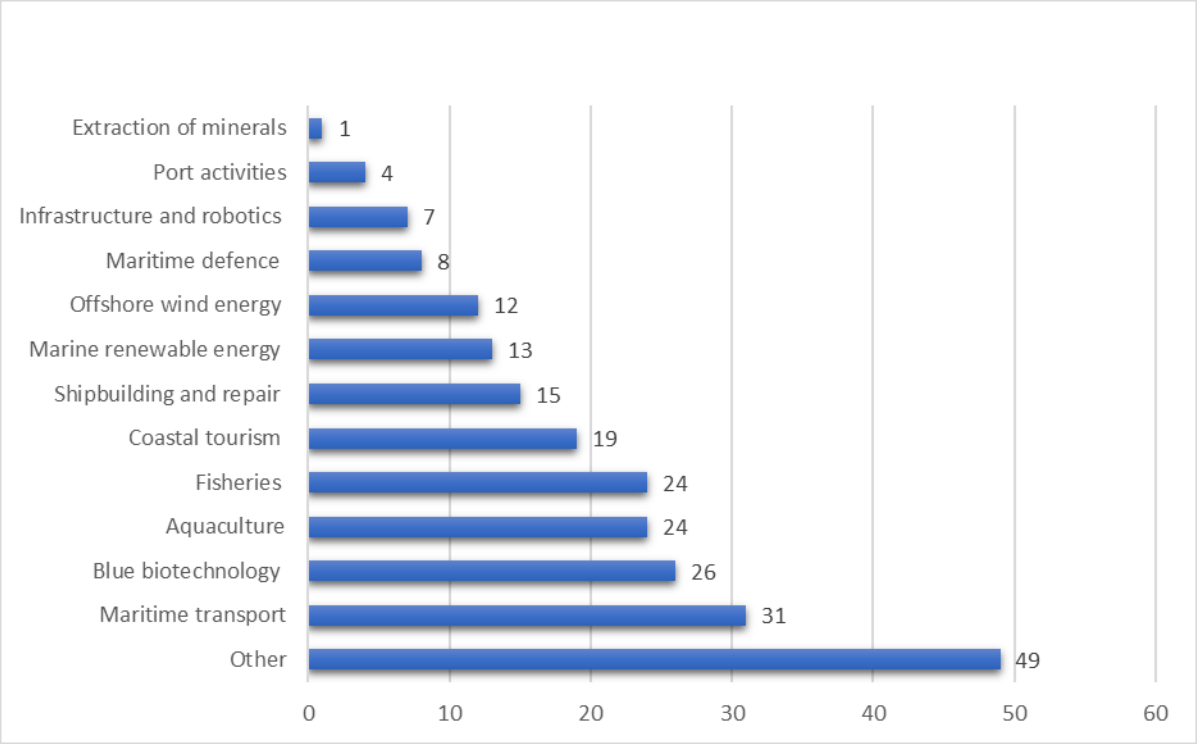
2. Identification

In this section of the survey, respondents were asked to indicate the sector that best describes their involvement in the blue economy, their years of experience in the sector, educational level, gender, country of residence, and the sea basin(s) their answers referred to.

Respondents by sector

Figure 1 illustrates the distribution of survey respondents across various sectors within the blue economy. With 49 respondents, "Other" is the largest group, indicating a significant number of participants whose roles do not fall neatly into predefined sector categories. On closer inspection, participants who identified themselves as "Other" mostly represent research in environmental conservation, or are affiliated with a maritime cluster.

Figure 1 Number of Respondents by Sector



The Maritime Transport sector is the second most represented, with 31 respondents, followed by Blue Biotechnology (26 respondents) and Aquaculture and Fisheries (24 respondents each).

Sectors such as Coastal Tourism (19 respondents) and Shipbuilding and Repair (15 respondents) also show moderate representation. Conversely, Port Activities (4 respondents) are underrepresented.

When interpreting the results of this analysis, it is important to consider the varying number of responses received from each sector, as illustrated in Figure 1. Sectors with fewer responses, such as Port Activities, may not provide as comprehensive insights as those with a higher number of responses, such as Maritime Transport or Blue Biotechnology. This difference in response volume should be kept in mind when drawing conclusions or comparing sectors, as it may influence the representativeness and depth of the findings.

At the same time, it is important to recognize that the number of individual responses per sector in the blue economy might be somewhat misleading. To ensure the survey results

provided a balanced perspective on the opinions of blue economy stakeholders, EU and national business associations were specifically targeted as respondents and asked to share the survey with their members. When a business association completed the survey, their response was recorded as a single entry. However, it can be argued that this single entry actually reflects the collective stance of a much larger group of companies.

Lastly, it should also be noted that no responses were received from the Desalination and the Oil & Gas sectors, whereas only one response was received from the Extraction of Minerals sector. While this response is valuable, it is believed that it is not sufficient to draw sector-specific conclusions. The reply is included in the broader analysis, but no sector-specific conclusions are inferred from it.

Years of experience

Over half of the respondents (55%) have more than a decade of experience in the blue economy, underscoring the maturity and expertise of the surveyed cohort. Early-career professionals (1–6 years of experience) account for 29% of participants, highlighting the sector's potential to attract younger talent. However, the low representation of new entrants (4%) indicates a possible gap in outreach or an ongoing challenge in attracting fresh talent to the blue economy.

Table 1 Years of experience in the blue economy sector

Years of Experience	Number of Respondents	Percentage
10+ years	129	55%
7–10 years	27	12%
4–6 years	36	15%
1–3 years	32	14%
Less than 1 year	9	4%

Education level

The high educational attainment of respondents is evident, with 83% holding either a master's degree or a PhD. This reflects the technical and interdisciplinary demands of blue economy roles, particularly in sectors like Marine Renewable Energy and Blue Biotechnology.

Table 2 Education level

Education Level	Number of Respondents	Percentage
Master's degree	106	45%
PhD	89	38%
Bachelor's degree	30	13%
Other	5	2%
High school diploma	3	1%

These figures indicate that the survey respondents are predominantly experienced professionals with advanced education levels, reflecting the technical and interdisciplinary nature of the blue economy.

Gender distribution

As far as the gender distribution is concerned, among the total of 233 respondents, male respondents outnumber female respondents across most sectors, with 58% of respondents identifying as male and 41% as female.

Table 3 Gender distribution

Gender	Number of Respondents	Percentage
Male	133	58%
Female	96	41%
Prefer not to say	4	1%

While this reflects historical trends in many blue economy sectors, the strong representation of women signals progress toward greater gender inclusivity, particularly in emerging fields like Blue Biotechnology and Coastal Tourism.

Top 5 countries

Table 4 Top 5 countries by respondent

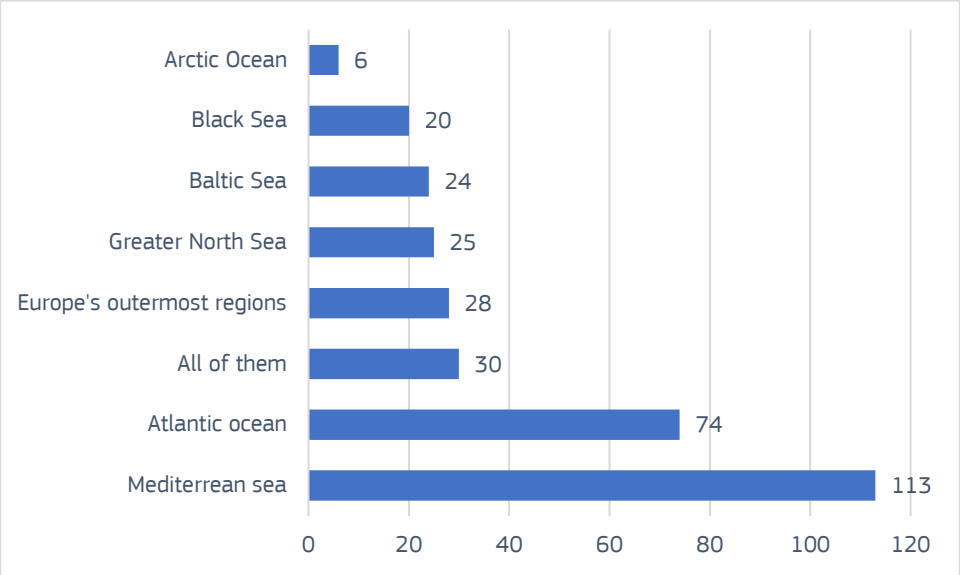
Country	Number of Respondents	Percentage
Spain	45	19%
Italy	32	14%
Greece	25	11%
France	18	8%
Portugal	18	8%

Spain leads with 44 respondents (19%), primarily from sectors such as Maritime Transport (8 respondents), Fisheries (8 respondents), Offshore Wind Energy, Blue Biotechnology, and Coastal Tourism (7, 6, and 5 respondents, respectively). Italy (14%) and Greece (11%) follow closely, with respondents mainly from Fisheries in Italy and Coastal Tourism and Maritime Transport in Greece.

In addition to EU Member States, the study included contributions from non-EU countries such as Turkey, South Africa, and Israel.

Sea basin

Figure 2 Number of responses referring to each sea basin



The Mediterranean Sea and the Atlantic Ocean stand out as the most frequently selected basins, with the Mediterranean Sea receiving the highest number of references, followed by the Atlantic Ocean.

Basins such as the Baltic Sea, Black Sea, and Greater North Sea received moderate levels of attention, and the Arctic Ocean was the least frequently referenced sea basin.

Additionally, the category "All of them" suggests a subset of respondents considered their answers applicable across all basins.

Please note that in the survey, respondents were allowed to select multiple sea basins to which their answers referred, and the majority chose more than one basin. As a result, it became challenging to attribute specific trends or insights exclusively to individual sea basins. This overlap in sea basin selection limits the ability to draw distinct, basin-specific conclusions. Therefore, the analysis in this report primarily focuses on sector-specific characteristics, which provide clearer and more actionable insights into the trends and priorities highlighted by the respondents.

3. Current skills profile and future needs

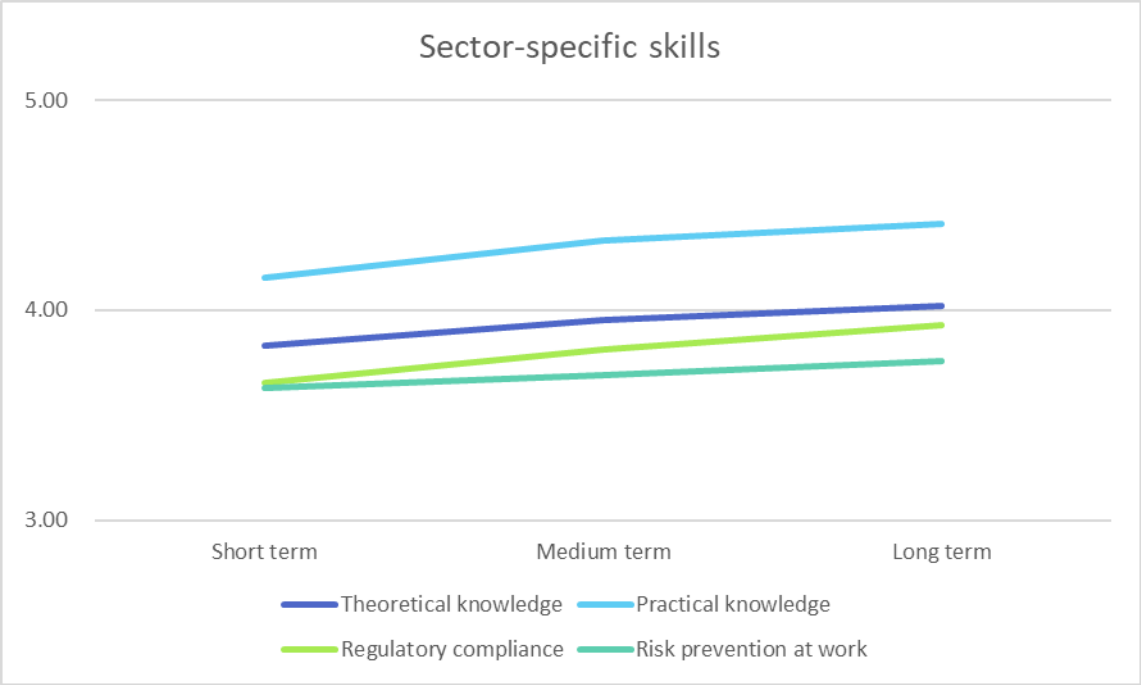
This section aimed to better understand the skills currently required by workers in the blue economy and those anticipated to be needed in the future. To facilitate the analysis, the skills have been grouped into five macro categories: sector-specific skills, digital skills, green skills, soft and cross-cutting skills, and business and management skills. Respondents were also given the opportunity to flag additional skills not included in the list through an open-ended question.

3.1 Sector-specific skills

Q11. Please rate the importance of the following skills that are specific to your sector for each time horizon. From 1 (not important) to 5 (very important).

Across all sectors, “practical knowledge” is identified as the most sought-after, consistently achieving the highest rankings in short-, medium-, and long-term evaluations, with an average score of 4.16 in the short term and 4.41 in the long term. In contrast, "risk prevention at work" is considered the least important skill set, receiving the lowest ratings. Nonetheless, it retains a relatively moderate score of 3.63 out of 5 in the short term, increasing slightly to 3.76 in the long term.

Figure 3 Sector-specific skills (average for all sectors)



The survey results reveal some sector-specific trends when comparing the importance of various skills over the short- and long-term:

- Theoretical knowledge:
    - **Aquaculture**: the importance of theoretical knowledge in aquaculture shows remarkable growth, with ratings increasing from 3.78 in the short term to 4.43 in the long term—a 15% rise, the highest across all sectors. This trend suggests a potential paradigm shift from traditionally family-run enterprises in certain Member States to more innovative, high-tech companies.
    - **Infrastructure and Robotics** and **Offshore Wind Energy**: respondents foresee a gradual decline in the value of theoretical knowledge in these sectors. Infrastructure and Robotics sees a decrease from 4.0 to 3.5 (-14%), while Offshore Wind Energy drops from 4.1 to 3.8 (-8%) over the same period.

- **Blue biotechnology**: this sector places the greatest emphasis on theoretical knowledge in the short term (4.28), maintaining a consistent rating in the long term, reflecting its stable importance.

- Practical knowledge
  - **Offshore Wind Energy** and **Shipbuilding and Repair:** both sectors anticipate an increased reliance on practical knowledge. Offshore Wind Energy ratings grow from 3.8 to 4.3 (+12%), while Shipbuilding and Repair experiences a similar trend, rising from 3.93 to 4.75 (+15%).
  - **Port activities:** practical knowledge holds consistently high value in this sector, with ratings improving slightly from 4.5 in the short term to 4.75 in the long term.

- Regulatory compliance
  - **Blue biotechnology:** regulatory compliance emerges as a critical factor, with its importance surging by 20%—from 3.28 in the short term to 4.08 in the long term, the largest increase among all sectors.
  - **Offshore Wind Energy and Port Activities:** conversely, these sectors predict a decrease in the importance of regulatory compliance. Offshore Wind Energy ratings decline from 4.0 to 3.7 (-8%), and Port Activities drop significantly from 4.5 to 3.25 (-28%).
  - **Infrastructure** and **Marine Renewable Energy:** both sectors assign relatively low priority to regulatory compliance, with Infrastructure remaining steady at 2.86, and Marine Renewable Energy experiencing a slight dip from 3.82 to 3.7.

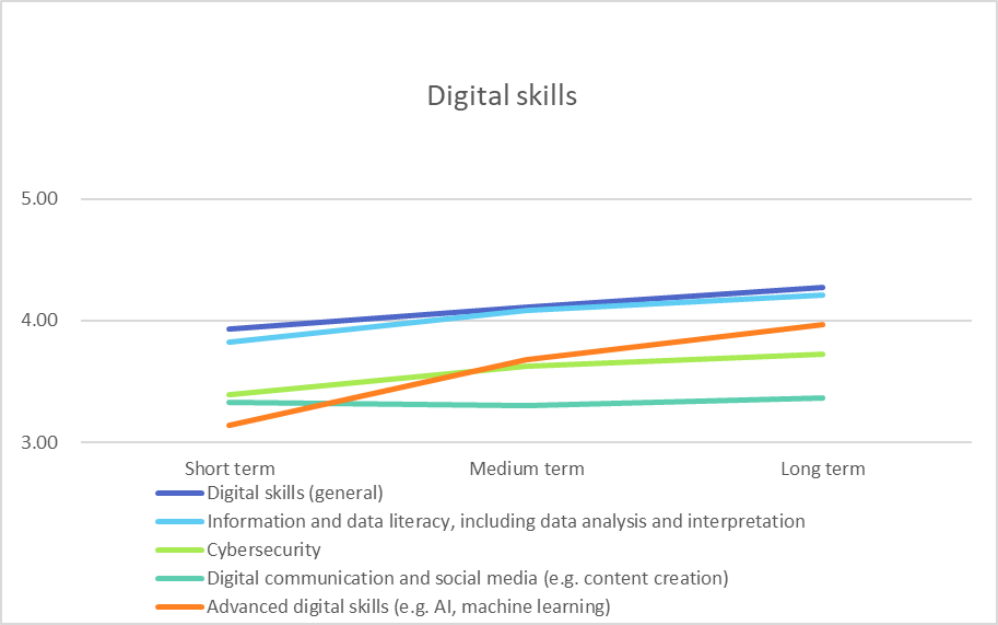
- Risk prevention at work:
  - **Infrastructure and Robotics:** the importance of risk prevention is expected to diminish, with ratings falling from 3.0 to 2.2 (-36%). Port Activities also project a decline, from 4.25 to 3.75 (-13%).
  - **Coastal Tourism** and **Shipbuilding and Repair**: these sectors consistently rate risk prevention highly. Coastal Tourism ratings rise from 4.06 to 4.24, and Shipbuilding and Repair follows suit, increasing from 4.0 to 4.31.
  - **Blue Biotechnology**: although rated lower compared to other sectors, the significance of risk prevention is expected to grow steadily, with ratings climbing from 2.83 to 3.27 over the long term.

3.2 Digital skills

Q12. Please rate the importance of the following digital skills for your sector for each time horizon. From 1 (not important) to 5 (very important)

The survey data highlights a clear trend: the importance of digital skills across all categories, but Digital Communication, grows significantly when analysed over the short, medium, and long term. This trend can signal the pivotal role of digital transformation in ensuring sectoral competitiveness and sustainability.

Figure 4 Digital skills (average for all sectors)



The survey results reveal some sector-specific trends when comparing the importance of various skills over the short- and long-term:

- General Digital Skills
    - **Shipbuilding and Repair and Offshore Wind Energy:** both sectors exhibit significant growth in the importance of general digital skills. Shipbuilding and Repair ratings rise from 3.29 in the short term to 4.29 in the long term (+23%), highlighting its potential for digital transformation. Offshore Wind Energy follows with an increase from 4.10 to 4.56 (+10%), reflecting growing adoption of digital tools to support sustainable energy production.
    - **Maritime Defence:** a slight decline is observed in the perceived importance of general digital skills in this sector, with ratings dropping from 3.57 to 3.29 (-9%).
    - **Coastal Tourism:** general digital skills remain consistently valued in this sector, with a modest increase in ratings from 4.18 to 4.41 (+5%).
    - **Maritime Transport, Port Activities and Coastal Tourism:** these sectors assign the highest importance to general digital skills both in the short and long term. Maritime Transport increases from 4.26 to 4.37 (+3%), Port Activities grow from 4.25 to 4.50 (+6%), and Coastal Tourism rises from 4.18 to 4.41 (+5%).

- Advanced digital skills
    - **Coastal Tourism:** this sector demonstrates the largest growth in advanced digital skills, with ratings surging from 2.94 in the short term to 4.06 in the long term (+28%).. **Maritime Transport and Shipbuilding and Repair** also display significant growth (+26% both).
    - **Marine Renewable Energy and Shipbuilding and Repair:** although initially rated as less critical, advanced digital skills gain importance, with Marine Renewable Energy increasing from 2.56 to 3.40 (+25%) and Shipbuilding and Repair rising from 2.85 to 3.86 (+26%).
    - **Port Activities:** this sector assigns the highest short-term rating to advanced digital skills (4.00), with ratings increasing further to 4.75 in the long term (+16%).
    - **Fisheries** and **Marine Renewable Energy**: these sectors rate advanced digital skills as less critical overall. Fisheries start at 2.85, growing to 3.84 (+13%), while Marine Renewable Energy begins at 2.56, rising to 3.40 (+25%).
- Information and data literacy
    - **Shipbuilding and Repair** and **Coastal Tourism:** both sectors show significant growth in the importance of information and data literacy. Shipbuilding and Repair ratings increase from 3.50 to 4.50 (+22%), while Coastal Tourism rises from 3.56 to 4.19 (+15%).
    - **Blue Biotechnology:** this sector consistently rates the importance of information and data literacy highly, with a slight increase from 4.20 to 4.28 (+2%).
    - **Port Activities** and **Infrastructure**: these sectors show gradual improvement in ratings from a baseline of 3.50 but remain lower in prioritisation compared to others..
- Digital communication
    - **Shipbuilding and Repair** and **Fisheries:** both sectors show moderate growth in the importance of digital communication skills, with Shipbuilding and Repair ratings increasing from 2.62 to 3.00 (+13%) and Fisheries rising from 3.05 to 3.37 (+9%).
    - **Coastal Tourism** and **Maritime Defence**: these sectors place the highest importance on digital communication skills in the short term with initial ratings of 4.19 and 4.00, respectively. However, Coastal Tourism experiences a slight decline (-2%), while Maritime Defense ratings remain stable over time.
    - **Marine Renewable Energy** and **Infrastructure and Robotics**: these sectors assign the lowest ratings to digital communication skills. Ratings in Infrastructure and Robotics remain unchanged at 2.67, while Marine Renewable Energy increases slightly from 2.78 to 3.00 (+7%).

- **Offshore Wind Energy:** this sector notes the most significant decline in the importance of digital communication skills, with ratings dropping from 3.20 to 2.78 (-15%).

- Cybersecurity
  - **Marine Renewable Energy**: although not initially rated as highly important, cybersecurity is expected to grow in importance within this sector, with ratings increasing from 3.00 to 3.70 (+19%, the largest increase among all sectors). In contrast, **Offshore Wind Energy** sees a slight decline in ratings, dropping from 3.60 to 3.44 (-5%).
  - **Port Activities** and **Maritime Transport**: these sectors assign the highest importance to cybersecurity. Port Activities maintain a perfect rating of 5.00, while Maritime Transport sees an increase from 3.81 to 4.30 (+11%).
  - **Blue Biotechnology**: this sector gives cybersecurity the lowest ratings, with scores increasing modestly from 2.83 in the short term to 3.12 in the long term.

3.3 Green skills

Q13. Please rate the importance of the following green skills for your sector for each time horizon. From 1 (not important) to 5 (very important).

Figure 4 illustrates the average importance of green skills—Environmental Sustainability Knowledge, Environmental Management and Conservation, and Circular Economy Principles—across all sectors over short, medium, and long-term timeframes.

Environmental Sustainability Knowledge is consistently rated as the most important, reflecting its foundational role in sustainability strategies. Environmental Management and Conservation also shows steady importance but slightly trails behind sustainability knowledge. Circular Economy Principles, while starting at a lower level, demonstrates the steepest growth, highlighting its rising significance, as sectors increasingly recognise the value of circular practices for long-term sustainability.

Overall, the trend suggests that all the three green skills are expected to gain importance in the long term, underscoring their critical role in achieving sustainable practices across sectors.

Figure 5 Green skills (average for all sectors)

The survey results reveal some sector-specific trends when comparing the importance of various skills over the short- and long-term:

- Environmental sustainability knowledge
    - **Infrastructure and Robotics**: this sector sees a remarkable rise in the importance of environmental sustainability knowledge, with ratings surging from 3.78 in the short term—the lowest across all sectors—to 4.60 in the long term (+26%).
    - **Coastal Tourism**: environmental sustainability knowledge also gains traction in this sector, where ratings increase significantly from 4.06 to 4.76 (+15%).
    - **Port Activities and Offshore Wind Energy**: while these sectors place high importance on sustainability knowledge in the short term (4.75 and 4.60, respectively), both experience a slight decline in the long term. Port Activities sees a 12% drop—the highest decline across all sectors—falling to 4.25, while Offshore Wind Energy decreases by 6% to 4.33.
    - **Blue Biotechnology**: this sector consistently values environmental sustainability knowledge highly, with a modest increase in importance from 4.33 to 4.65 (+7%).
- Environmental management and conservation
    - **Infrastructure and Robotics and Shipbuilding and Repair**: both sectors experience substantial growth in the importance of environmental management and conservation skills. Infrastructure and Robotics sees ratings

leap from 3.20 to 4.80 (+33%), marking the largest growth across all sectors, while Shipbuilding and Repair rises from 3.08 to 3.92 (+22%)..

- **Port Activities and Fisheries**: these sectors consistently prioritize environmental management. Port Activities achieves a perfect long-term rating of 5.0, up from 4.50, while Fisheries improves from 4.16 to 4.67.
- **Offshore Wind Energy**: this sector recognizes the importance of environmental management and conservation skills in the short term, with a relatively high rating of 4.20, but then shows a slight decline to 4.00, indicating reduced focus compared to other sectors. In contrast, **Marine Renewable Energy** demonstrates an opposite trend, starting with a short-term rating of 3.60 and growing modestly to 4.10 in the long term.

- Circular economy principles
  - **Infrastructure and Robotics, Shipbuilding and Repair, Coastal Tourism:** all these sectors project growth in the importance of circular economy principles. Infrastructure and Robotics demonstrates the largest growth, starting at 3.20 and increasing to 4.20 in the long term (+24%). Shipbuilding and Repair shows an identical growth rate of +24%, starting at 3.08 and reaching 4.08. The importance of this skill in the Coastal Tourism sector is expected to rise by 16%, from 3.78 in the short term to 4.50 in the long term.
  - **Blue Biotechnology**: this sector leads the way in valuing circular economy principles, starting with the highest rating across all sectors at 4.63 in the short term and maintaining this leadership with a slight increase to 4.75 in the long term.

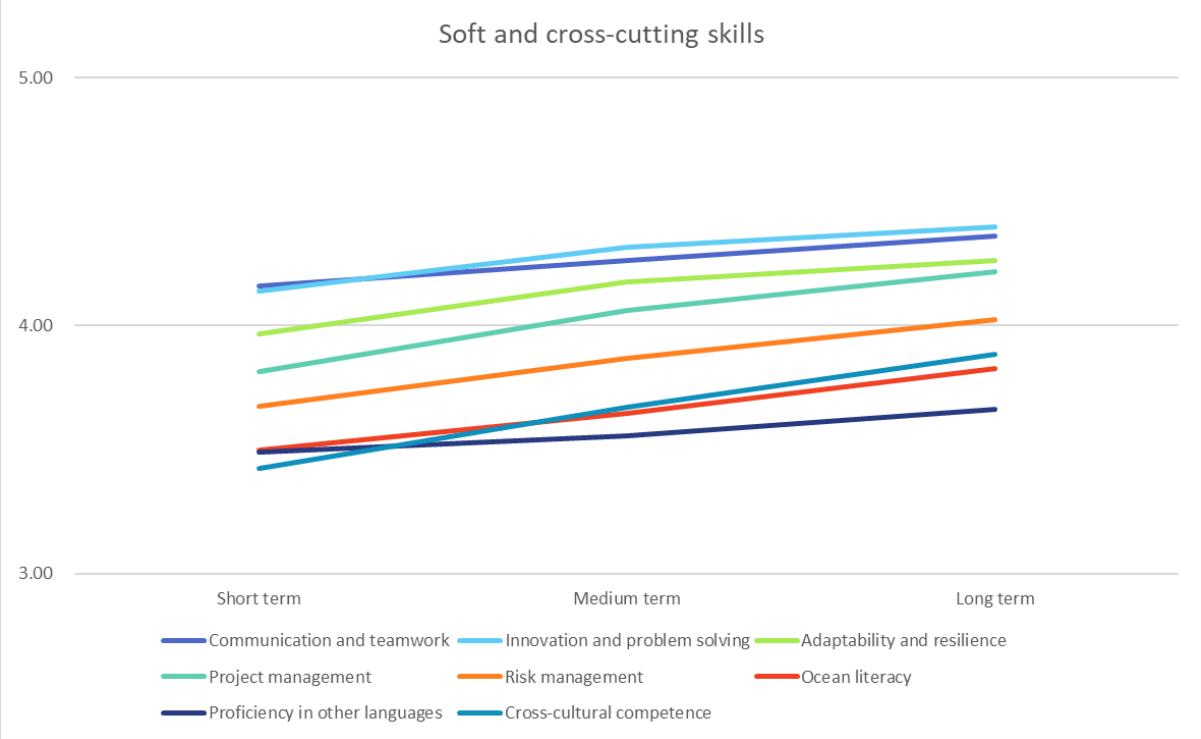
3.4 Soft and cross-cutting skills

Q14. Please rate the importance of the following soft and cross-cutting skills for your sector for each time horizon. From 1 (not important) to 5 (very important).

Figure 5 highlights the average importance of Soft and Cross-Cutting Skills—such as Communication and Teamwork, Innovation and Problem Solving, and Ocean Literacy—across sectors over short, medium, and long-term timeframes.
Communication and Teamwork and Innovation and Problem Solving consistently show the highest importance across all timeframes. Adaptability and Resilience and Project Management demonstrate steady growth. Ocean Literacy and Proficiency in Other Languages, while starting with relatively lower importance, show noticeable growth over time. Cross-Cultural Competence, though slightly lower, follows a similar trend.

Overall, the figure indicates that all soft skills will continue to gain importance in the long term, highlighting their integral role in fostering collaboration, innovation, and adaptability across sectors.

Figure 6 Soft and cross-cutting skills (average for all sectors)



The survey results reveal some sector-specific trends when comparing the importance of various skills over the short- and long-term:

- **Project Management**
    - **Infrastructure and Robotics** and **Aquaculture**: both sectors show strong growth in the importance of project management. Infrastructure and Robotics leads with a 16% increase, climbing from 3.50 to 4.17, reflecting its growing reliance on structured management. Aquaculture follows with a 13% growth...
    - **Port Activities and Marine Renewable Energy**: these sectors remain stable, highlighting their ongoing reliance on project management. Port Activities maintains a steady rating of 4.25, while Marine Renewable Energy holds consistent at 4.00.
    - **Blue Biotechnology**: this sector prioritizes project management, starting strong at 4.08 and growing to 4.64 (+12%), underscoring its increasing focus on complex project execution.
    - **Maritime Defence**, **Fisheries**, and **Infrastructure and Robotics**: starting with lower scores, these sectors show varied growth. Maritime Defence improves modestly from 3.43 to 3.57 (+4%), Fisheries grows steadily from 3.60 to 4.12 (+13%), and Infrastructure and Robotics demonstrates the highest growth potential (+16%).

- **Communication and teamwork**
    - **Coastal Tourism** and **Fisheries**: both sectors demonstrate strong growth in the importance of communication and teamwork, with a trend of +8%, reflecting their increasing emphasis on collaboration.
    - **Aquaculture**: this sector leads in prioritisation, starting with the highest short-term rating (4.64) and rising to 4.90, showcasing its robust focus on teamwork.
    - **Maritime Transport** and **Infrastructure and Robotics**: Maritime Transport scores consistently high, holding steady at 4.54, while Infrastructure and Robotics sees a slight decline, dropping from 4.50 to 4.33 (-4%).
    - **Port Activities** and **Maritime Defence**: these sectors place less emphasis on communication and teamwork. Port Activities holds the lowest rating (3.75) with no growth, while Maritime Defence rises modestly from 3.86 to 4.00 (+4%).
- **Innovation and Problem Solving**
    - **Aquaculture**, **Fisheries**, and **Coastal Tourism**: these sectors show strong growth in the importance of innovation and problem-solving. Aquaculture leads with a 12% increase (4.26 to 4.82), followed by Fisheries (+10%) and Coastal Tourism (+8%).
    - **Port Activities** and **Blue Biotechnology**: both sectors demonstrate high prioritisation of the innovation and problem-solving skills. Port Activities scores the highest in the long term, increasing from 4.75 to 5.00 (+5%). Blue Biotechnology also prioritizes innovation, starting at 4.60, but shows a slight decline to 4.56 (-1%).
    - **Infrastructure and Robotics**: this sector sees the largest decline, dropping from 4.50 to 4.33 (-4%), reflecting a reduced emphasis on innovation.
    - **Marine Renewable Energy**: starts at 3.89 and grows modestly to 4.00 (+3%), indicating slower adoption.
- **Adaptability and Resilience**
    - **Infrastructure and Robotics**, **Fisheries**, and **Marine Renewable Energy:** significant growth is evident, with Infrastructure and Robotics leading at +23% (3.40 to 4.40), followed by Fisheries (+15%) and Marine Renewable Energy (+12%).
    - **Port Activities** and **Maritime Transport**: these sectors maintain high importance, with Port Activities scoring the highest across all timeframes (4.75) and no growth observed. Maritime Transport starts high (4.26) but experiences a slight decline (-3%).

- **Marine Renewable Energy** and **Infrastructure and Robotics**: these sectors begin with relatively low importance but show steady growth. Marine Renewable Energy starts at 3.33 and grows to 3.80 (+12%), while Infrastructure and Robotics starts at 3.40 but grows significantly to 4.40 (+23%).

- **Cross-Cultural Competence**
  - **Infrastructure and Robotics**, **Fisheries**, **Marine Renewable Energy**, and **Shipbuilding and Repair**:all these sectors show significant growth in the importance of cross-cultural competence. Infrastructure and Robotics demonstrates the highest growth, with a 28% increase (2.60 to 3.60). Fisheries follows with +20% growth (3.26 to 4.06), while Marine Renewable Energy and Shipbuilding and Repair both grow by +17% (2.67 to 3.20 and 3.07 to 3.71, respectively).
  - **Maritime Defence** and **Maritime Transport**: modest growth is observed. Maritime Defence ends with the highest long-term rating (4.29), while Maritime Transport grows by +6% (3.89 to 4.12)..
  - **Offshore Wind Energy** and **Port Activities**: both sectors remain stable, showing no growth over time, with scores of 3.11 and 3.75, respectively.

- **Risk Management**
  - **Infrastructure and Robotics**, **Fisheries**, and **Blue Biotechnology**:these sectors demonstrate strong growth in risk management. Infrastructure and Robotics leads with a +29% increase (3.00 to 4.20), followed by Fisheries (+20%) and Blue Biotechnology (+11%).
  - **Port Activities**: this sector maintains consistently high importance remaining stable at 3.75 across all timeframes.
  - **Offshore Wind Energy** and **Maritime Transport**: both sectors show minimal growth in the importance of risk management skills, with Offshore Wind Energy increasing from 4.00 to 4.11 (+3%) and Maritime Transport rising from 3.96 to 4.23 (+7%).

- **Ocean Literacy**
  - **Fisheries**, **Maritime Transport**, and **Shipbuilding and Repair**: these sectors show strong growth in the importance of ocean literacy. Fisheries demonstrates the highest growth (+17%), increasing from 3.42 in the short term to 4.11 in the long term. Maritime Transport follows with a +16% increase, reflecting the sector's growing awareness of ocean literacy. Shipbuilding and Repair grows steadily by +11%, emphasising its rising importance.

- **Port Activities** and **Coastal Tourism** rank highest in prioritisation. Port Activities scores the highest in the short term (4.25) and remains stable over time. Coastal Tourism starts at 4.00 and grows to 4.29 (+7%), reflecting consistent high prioritisation of ocean literacy.
- **Infrastructure and Robotics**, **Maritime Transport**, and **Shipbuilding and Repair**: these sectors have lower starting scores but show improvement. Infrastructure and Robotics starts with the lowest score (2.40) and grows modestly to 2.60 (+8%). Maritime Transport and Shipbuilding and Repair also start low (2.92 and 2.93, respectively) but demonstrate significant growth in importance.

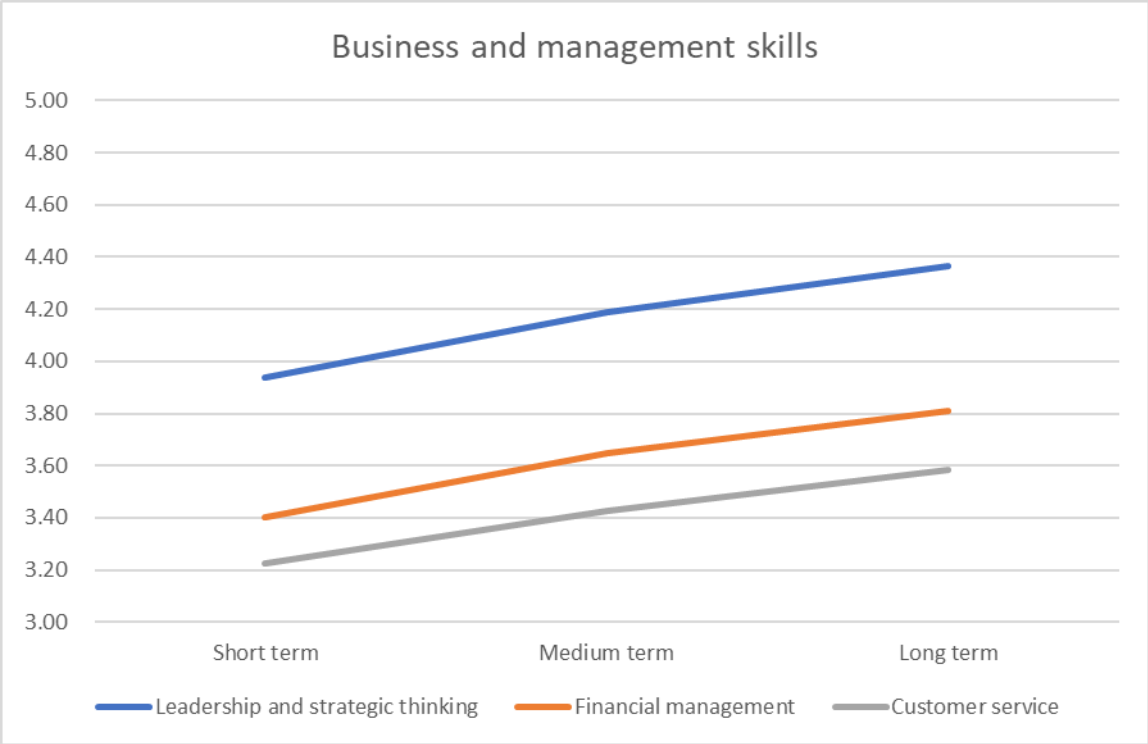
- **Proficiency in Other Languages**
  - **Fisheries** and **Infrastructure and Robotics**: these sectors demonstrate strong growth. Fisheries increases by +11% (3.35 to 3.78), and Infrastructure and Robotics grows by +9%, reflecting its increasing adoption of this skill.
  - **Port Activities**: shows the sharpest decline, dropping by -14% from 4.00 in the short term to 3.50 in the long term, indicating reduced prioritisation of language skills.
  - **Coastal Tourism**, and **Maritime Defence**: rank high in short-term prioritisation. Coastal Tourism begins strong at 3.88 and grows slightly to 3.94 (+1%), while Maritime Defence maintains a relatively high rating, starting at 3.71 and ending at 3.86.
  - **Infrastructure and Robotics** and **Marine Renewable Energy**: these sectors start with lower importance but show modest growth. Infrastructure and Robotics begins at 2.50 and grows to 2.78, while Marine Renewable Energy starts at 2.78 and increases to 3.00.

3.5 Business and managerial skills

Q15. Please rate the importance of the following business and managerial skills for your sector for each time horizon. From 1 (not important) to 5 (very important).

Figure 6 illustrates the overall importance of three key business and managerial skills—Leadership and Strategic Thinking, Financial Management, and Customer Service—across all sectors of the blue economy. It highlights how the perceived importance of each skill evolves over the short, medium, and long term. Notably, Leadership and Strategic Thinking consistently ranks the highest, showing steady growth across timeframes. Financial Management follows a similar upward trajectory, reflecting its increasing significance. Customer Service, while scoring lower, also demonstrates a gradual rise in importance, emphasising its growing relevance in future-oriented sectoral strategies.

Figure 7 Business and management skills (average for all sectors)



The survey results reveal some sector-specific trends when comparing the importance of various skills over the short- and long-term:

- **Financial management**
    - **Infrastructure and Robotics**, **Blue Biotechnology**, and **Coastal Tourism**: these sectors exhibit strong growth in the importance of financial management. Infrastructure and Robotics leads with a 33% increase, rising from 2.40 to 3.60, showcasing a growing emphasis on fiscal expertise. Blue Biotechnology follows with a 19% rise (3.38 to 4.17), while Coastal Tourism increases by 15% (3.50 to 4.13), reflecting the growing need for financial acumen in tourism management.
    - **Marine Renewable Energy** and **Maritime Defence**: these sectors display a declining emphasis on financial management. Marine Renewable Energy decreases by 8% (4.11 to 3.80), while Maritime Defence experiences a slight drop of 5% (3.14 to 3.00).
    - **Port Activities**, **Shipbuilding and Repair**: both sectors remain stable over time. Port Activities maintains the highest rating (4.25), emphasising its critical reliance on financial management, while Shipbuilding and Repair shows no significant growth.
    - **Aquaculture** and **Blue Biotechnology**: these sectors rank high in long-term prioritisation. Aquaculture achieves a score of 4.18, closely followed by Blue Biotechnology at 4.17, underlining the importance of financial skills in their operations.

- **Infrastructure and Robotics** and **Maritime Defence**: these sectors start with the lowest scores but show varying trends. Infrastructure and Robotics grows substantially from 2.40 to 3.60 (+33%), while Maritime Defence stagnates at a modest 3.00.

- **Customer service**
  - **Infrastructure and Robotics**, **Blue Biotechnology**, and **Aquaculture**: these sectors demonstrate the strongest growth in customer service importance. Infrastructure and Robotics achieves the highest long-term score of 5.00 (+20%), indicating a transformative focus on customer relations. Blue Biotechnology follows with a 16% rise (3.09 to 3.70), and Aquaculture grows by 12% (2.82 to 3.20), reflecting enhanced customer engagement efforts.
  - **Offshore Wind Energy:** this sector sees the sharpest decline, dropping by 9% (2.90 to 2.67), signalling reduced prioritisation of customer-facing roles.
  - **Port Activities**, **Coastal Tourism**, and **Maritime Transport**: these sectors rank high in customer service prioritisation. Port Activities maintains a steady 4.25, Coastal Tourism reaches 4.13, and Maritime Transport follows closely at 4.08, showcasing their sustained focus on customer satisfaction.
  - **Fisheries** and **Offshore Wind Energy**: these sectors have the lowest long-term scores. Fisheries grows modestly from 2.75 to 3.06, while Offshore Wind Energy declines to 2.67, marking the lowest score and largest decrease (-9%)..

- **Leadership and strategic thinking**
  - **Infrastructure and Robotics**: this sector shows the highest growth in leadership and strategic thinking, rising by 23% (3.40 to 4.40). This reflects its increasing recognition of leadership's role in navigating complex projects and innovation.
  - **Port Activities**, **Marine Renewable Energy**, and **Aquaculture**: these sectors rank highest in long-term prioritisation. Port Activities leads with a consistent 4.75, followed by Aquaculture at 4.50 and Marine Renewable Energy at 4.40. Their high scores underscore the critical importance of strategic thinking in these industries.
  - **Coastal Tourism**, **Fisheries**, **Maritime Defence**, and **Offshore Wind Energy**: these sectors start with lower scores but show significant improvement over time. Coastal Tourism grows from 3.59 to 4.18 (+16%), Fisheries rises from 3.79 to 4.39 (+16%), and Maritime Defence and Offshore Wind Energy grow to 4.14 and 4.11, respectively, showcasing a steady focus on leadership development.

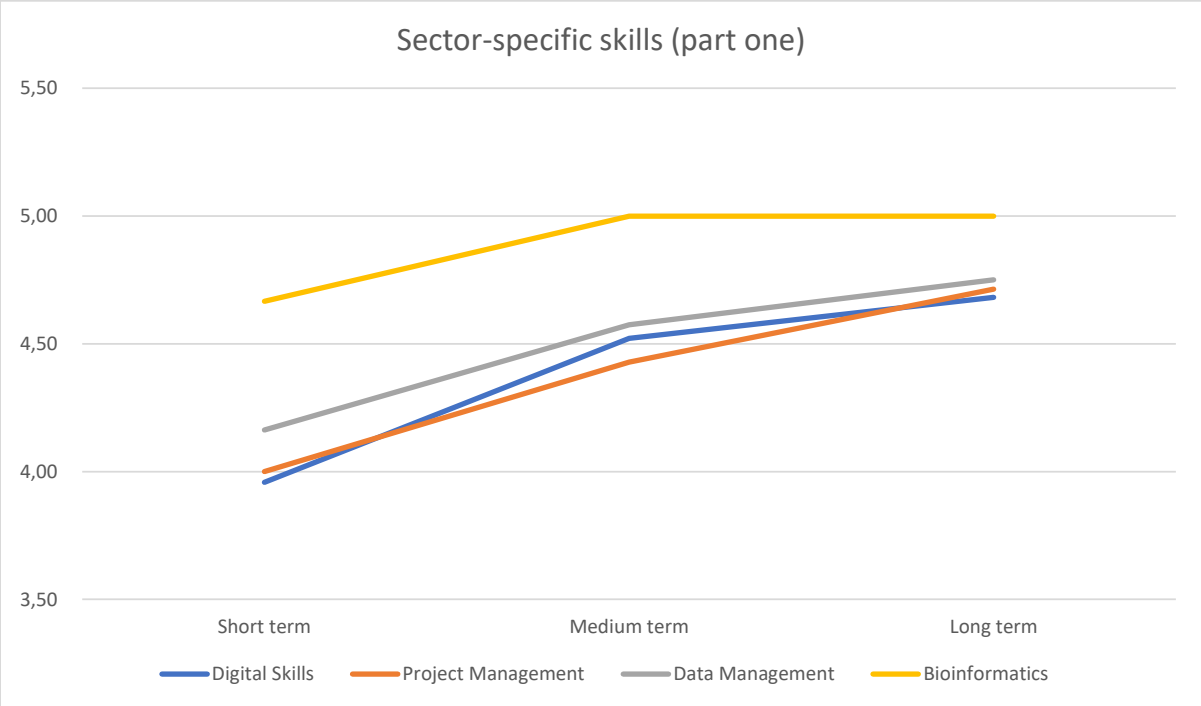
3.6 Open questions

Q16. Please list up to 5 technical skills that are specific to your sector and rate their importance for each time horizon. From 1 (not important) to 5 (very important).

Respondents were also asked to identify sector-specific skills, so as to have an overview of what jobs and roles are in the highest demand in each sector of the blue economy. Out of the 233 total participants, 156 of them indicated at least a sector-specific skills. Moreover, 37% of total participants listed up at least 5 sector-specific skills. The majority of them were more detailed and specialised than the categories provided in the survey. However, for consistency, they were broadly classified within the overarching skill categories outlined, offering a more nuanced perspective on the skills required across different sectors of the blue economy.

The analysis of technical skills across sectors reveals several emerging priorities that are consistently rated as important (average rating > 4.0) and frequently mentioned. These skills underscore the growing focus on innovation, efficiency, and sustainability in the blue economy and, for this reason, their importance over time, according to the respondents, generally grows. Key findings include:

Figure 8 Sector-specific skills, part one (average for all sectors)



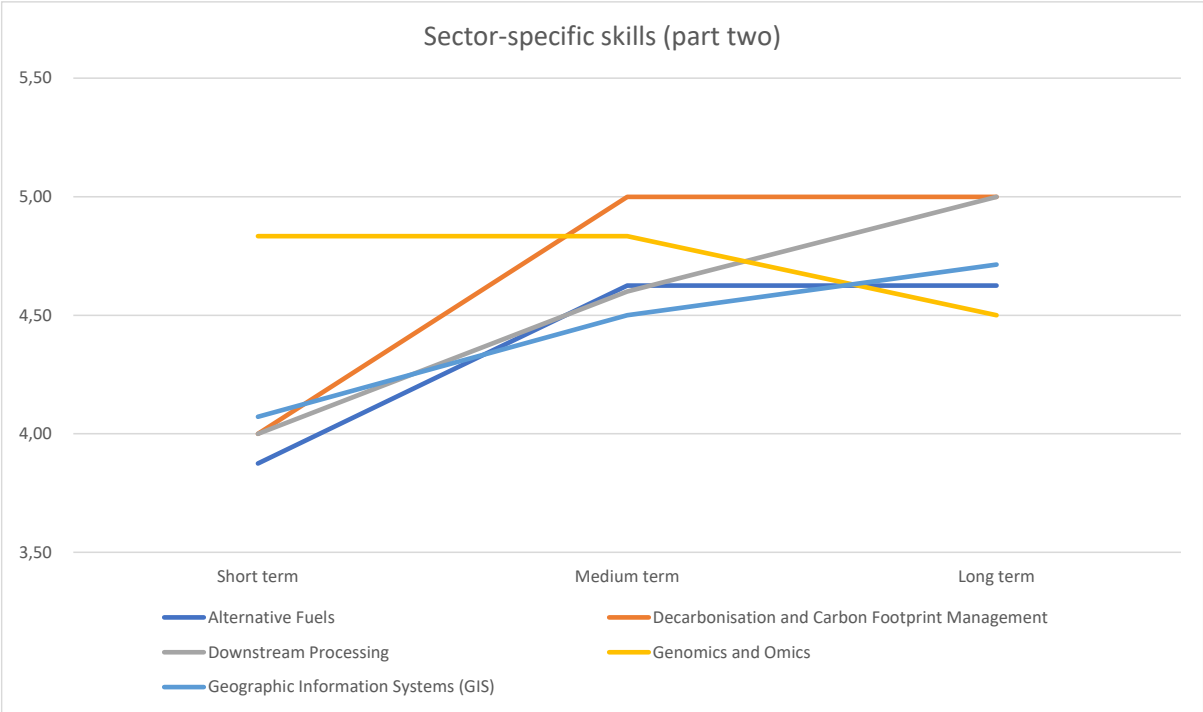
- **Digital Skills**. Digital skills are universally recognised as critical for all sectors. This reflects the increasing reliance on automation, data integration, and technological advancement to improve operational efficiency and decision-making. This trend is supported by the data, which shows the importance of digital skills growing over time, from a rating of 3.96 in the short term to 4.68 in the long term.

- **Project Management**. Effective project management remains a cornerstone for industries requiring streamlined operations and resource optimisation. This skill is particularly relevant in infrastructure, tourism, and energy sectors, as evidenced by ratings of 4.00 in the short term, 4.43 in the medium term, and 4.71 in the long term.

- **Data Management**. The ability to manage and analyse data is vital for sectors leveraging data-driven decision-making, it was cited 82 times and showed consistent growth in importance, from 4.16 in the short term to 4.75 in the long term. This includes Marine Renewable Energy and Maritime Transport operations.

- **Bioinformatics**. This specialised skill reflects the importance of leveraging biological data, particularly in Blue Biotechnology and marine science sectors. Its consistent high ratings—4.67 in the short term and a perfect 5.00 in both the medium and long term—highlight the growing integration of advanced computing in research and application.

Figure 9 Sector-specific skills, part one (average for all sectors)



- **Downstream Processing**. This skill is critical in sectors such as Marine Renewable Energy and Blue Biotechnology, focusing on refining raw outputs into usable products. Its high rating, 4.00 in the short term, 4.60 in the medium term, and 5.00 in the long term—underscores its growing importance in achieving efficiency and sustainability goals.

- **Decarbonisation and Carbon Footprint Management**. This skill is particularly significant in the Maritime Transport sector, reflecting a growing industry-wide commitment to sustainability and environmental responsibility. Its ratings—4.00 in the short term and a perfect 5.00 in both the medium and long term—indicate a strong focus on greener operations in alignment with global climate goals.

- **Genomics and Omics**. Skills related to genomics, integrative omics, and genome mining were cited primarily within the Blue Biotechnology sector. This underscores the importance of advanced biological and computational techniques for innovation in biotechnology and marine sciences. With a high initial rating of 4.83 in both the short and medium terms, and a slight dip to 4.50 in the long term, this skill remains vital for advancing biotechnology and marine sciences.

- **Geographic Information Systems (GIS)**. Proficiency in GIS is essential across diverse sectors such as Aquaculture, Fisheries and Offshore Wind Energy. Its ratings—4.07 in the short term, 4.50 in the medium term, and 4.71 in the long term—s reflects the critical role of spatial data analysis in resource management, environmental monitoring, and infrastructure planning.

- **Alternative Fuels**. The need for expertise in alternative fuels was heavily emphasised, particularly within Marine Renewable Energy and Maritime Transport. Skills related to the production, application, and bunkering of new fuel types (e.g., methanol, LNG, and ammonia) reflect the industry's transition to sustainable energy sources to reduce carbon emissions. The data shows an initial rating of 3.88 in the short term, which rises significantly to 4.63 in both the medium and long terms

Q17. Are there any additional skills not previously mentioned that you would like to rate in terms of their importance for your sector across different time horizons? Please list up to 5 skills that you consider important for your sector and are not included in the list above.

The survey results underscore the diversity of skills perceived as important across sectors, with a strong emphasis on adaptability, technological advancement, and strategic innovation. Temporal trends indicate that while immediate priorities focus on operational competencies, there is a clear shift toward long-term strategic capabilities, particularly those driven by technology and innovation.

Respondents identified a wide variety of skills, ranging from technical competencies to interpersonal and strategic abilities:

- **Frequently mentioned skills**. Skills like Application Development, Adaptability to Change, Technological Development, and Personal Skills appeared repeatedly across responses. These skills reflect a blend of immediate technical needs and long-term strategic priorities. Adaptability to Change emerged as a central theme, indicating its critical role in navigating dynamic and evolving challenges.

- **Less Frequently Mentioned but Specialised Skills**. Niche skills like Blockchain for Supply Chain Traceability, Recirculating Aquaculture Systems, and Genetic Improvement were highlighted by specific respondents. Although these skills were less frequently mentioned, their ratings suggest they are highly valued in specialised contexts.

Q18. What new job roles or occupations do you foresee emerging in your sector over the next decade?

The responses to the survey question on emerging job roles and occupations reveal a transformative shift across sectors, driven by advancements in technology, sustainability imperatives, and evolving industry challenges. The anticipated roles reflect the growing need for multidisciplinary expertise and innovative approaches to address future demands in the blue economy. Below are the key trends identified:

- **Technology-Driven Roles**
    - Across multiple sectors, there is a clear trend toward roles requiring advanced technological expertise. Examples include **AI Specialists, Data Scientists, Drone Operators, and Cybersecurity Analysts**. These roles reflect the growing reliance on automation, data analytics, and digitalisation in traditional and emerging industries.
    - Specific to sectors like offshore energy, **Digital Twin Engineers, Autonomous Ship Operators, and Robotics Specialists** will play vital roles in optimising performance and ensuring safety.
- **Sustainability-Focused Occupations**:
    - Many sectors anticipate an increase in roles dedicated to sustainability and environmental management. Titles like **Sustainability Manager, Circular Economy Specialist, and Marine Conservation Specialists** demonstrate a heightened emphasis on eco-friendly practices, recycling, and reducing ecological impacts.
    - The **Blue Carbon Specialist and Climate Resilience Advisor** roles highlight the integration of environmental economics with marine conservation efforts.
- **Sector-Specific Specialisations**:
    - In **aquaculture**, roles such as **Seaweed and Shellfish Farmers, Hatchery Technicians, and Sustainability Officers** will address production optimisation and ecosystem management.
    - **Blue biotechnology** sectors foresee demand for **Marine Biotechnologists, Waste Valorisation Experts, and Biorefinery Specialists**, focusing on deriving value from marine resources while maintaining ecological balance.
- **Emerging Cross-Sector Roles**. Many roles are interdisciplinary, combining expertise in multiple domains. For example:
    - **Energy Transition Managers** will oversee shifts to alternative fuels and renewable energy sources across maritime sectors.
    - **Marine Spatial Planners** will balance renewable energy projects, aquaculture, and conservation in coastal and offshore spaces.

- **Knowledge Brokers** will facilitate collaboration across academic, industrial, and policy domains to enhance innovation.

- **Infrastructure and Digital Transformation**:
  - Job roles like **Smart Ship Engineers, Augmented Reality Specialists**, and **Modular Shipbuilding Engineers** highlight the integration of cutting-edge technology in maritime infrastructure.
  - Positions such as **Digital Logistics Coordinators and Automation Engineers** in Maritime Transport and Port Activities reflect the growing emphasis on operational efficiency through digital solutions.
- **Challenges driving workforce needs**. Several sectors highlighted the need for jobs that address specific challenges. For example:
  - **Fishing Tourism Guides and Environmental Awareness Trainers** in Coastal Tourism aim to blend economic growth with public engagement and conservation.
  - **Waste Management Experts and Decommissioning Specialists** in Offshore Wind and Marine Renewable Energy focus on sustainable lifecycle management of installations.
- **Strategic Leadership and Policy Roles**:
  - High-level roles like **Blue Economy Policy Advisors** and **Ocean Governance Specialists** will drive policymaking and international collaboration.
  - Strategic roles such as **Sustainable Fisheries Managers and Marine Renewable Energy Planners** indicate a growing intersection of technology, policy, and sustainability.

4. Education, training and re-skilling/upskilling programmes

This section aimed to better understand the current state of education and training programmes within the blue economy and its perception, while also integrating the Contractor's ongoing efforts to map education, training, and upskilling programmes related to blue skills.

General findings

The findings of this section reveal that 58% of respondents are aware of existing blue skills training programmes, with sectors like **Maritime Transport, Fisheries, and Aquaculture** showing particularly high awareness rates. Participation in blue skills training has been widespread, with 63% of respondents reporting involvement in the past two years. **Workshops, webinars, and collaborative projects** were the most common training formats, while apprenticeships and mentoring programmes were less utilised, despite being

recognised as key areas for improvement. A notable gap between theoretical instruction and practical, hands-on training was identified, with respondents calling for more real-world application in blue skills curricula. Participants also highlighted several areas for improvement, including **enhanced collaboration between industry and education providers**, more **hands-on learning opportunities**, and **curricula better aligned with industry needs**. The integration of new technologies, such as Virtual and Augmented Reality, was also suggested as a way to partially replicate hands-on fieldwork, as these technologies can simulate realistic scenarios without the associated risks. Additionally, innovative training approaches such as gamification, simulation-based learning, and blended learning were proposed to make blue skills training more engaging and accessible.

Q19. Are you aware of any education, training or upskilling programmes related to blue skills in your region/country?

Out of the total 233 respondents, 136 or 58% indicated awareness of education, training, or upskilling programmes related to blue skills, while 90 or 39% replied negatively. The remainder, 7 or 3%, did not provide an answer.

Table 5

	Number of respondents	Percentage
Yes	136	58%
No	90	39%
Not answered	7	3%

Sector-specific results, for comparable sectors with a similar response rate, show that industries like **Maritime Transport, Fisheries, and Aquaculture** have similar awareness rates, averaging 61% of stakeholders who know about relevant programmes. By contrast, **Blue Biotechnology** demonstrated a lower awareness rate of 46%.

As per mapping of blue skills training programmes, based on the 119 respondents (88% of 136), approximately 30% of the total indicated link or training programme which were already covered by the internal database. Keeping in mind that some replies were repeated, while others referred to extra-EU initiatives and furthermore were too generic, the remainder projects have been included. Overall, 60 new training/upskilling programmes were added to the database.

Here below is a small extract with five interesting projects:

Table 6 Newly mapped training and upskilling programmes

Name of the project	Country	Scope
Specialised Advanced Master Marine Renewable energies expert	France	Giving students the opportunity to pursue careers as project managers/programme directors within the marine renewable energy (MRE) sector; Boosting the understanding of marine energy resources; Providing dual specialisation for becoming an expert in MRE

Name of the project	Country	Scope
Erasmus Mundus Joint Masters Degree in Aquaculture, Environment and Society	Greece	Providing practical and theoretical expertise for a successful career in aquaculture; Carrying out industry-led research projects and internships
Masters of the Sea	Italy	Recruiting, training and employing workers in the shipbuilding industry; Investing in technical expertise for direct employment at Fincantieri's shipyards across Northern, Central, and Southern Italy
FOR-MAR	Portugal	Enhancing human resources in maritime activities; Developing skills and career advancement of seafarers; Validating skills developed in learning environments for maritime professionals through examinations
Wind – Mine of Opportunities	Poland	Training miners in the operation and maintenance of onshore wind farm installations; Providing training sessions (BST Onshore, Fire Awareness, Working at Heights and First Aid)

Q20. Have you or your organisation participated in or provided any blue skills training programmes in the last two years?

Participation in blue skills training programmes was widespread, with 146 respondents, or 63%, indicating that they or their organisations had participated in or provided training in the last two years. The rate was particularly high, of 68%, among **academia**, the most represented group in the survey, and **civil society** stakeholders. However, 79 or 34% respondents stated they had not participated, while the remainder 8 or 4% provided no response.

Table 7

	Number of respondents	Percentage
Yes	146	63%
No	79	34%
Not answered	8	4%

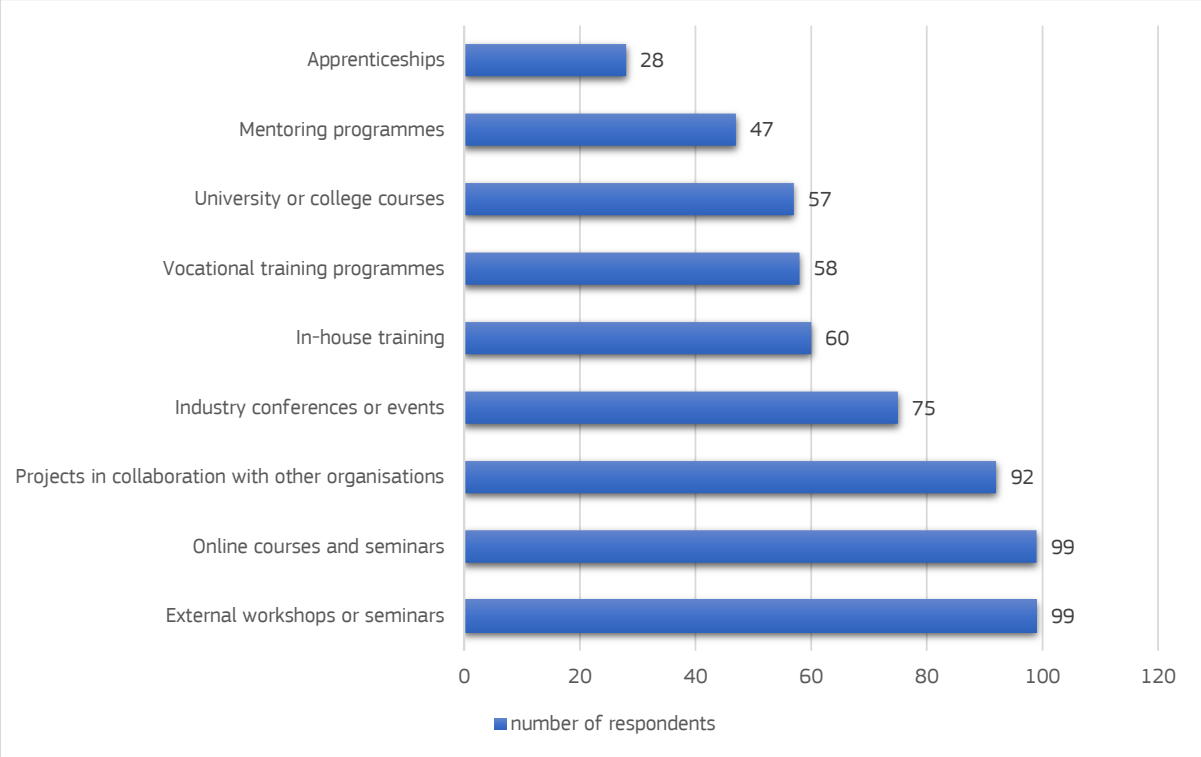
Q20a. What types of training programmes has your organisation participated in or provided?

Insights from those who participated or organised training programmes revealed that **workshops and webinars** were the most utilised format, having 42% of respondents participated in them, highlighting a preference for flexible and accessible learning opportunities. **Collaborative projects** followed closely at 39%.

Notably, **apprenticeships and mentoring programmes** were underutilised, with only 12% and 20% of respondents, respectively, indicating their involvement in such initiatives. This

aligns with other findings, for Q22 and Q23, that emphasise the demand for **hands-on fieldwork** and **mentoring programmes** as desirable improvements to blue skills training.

Figure 10 Type of training programmes

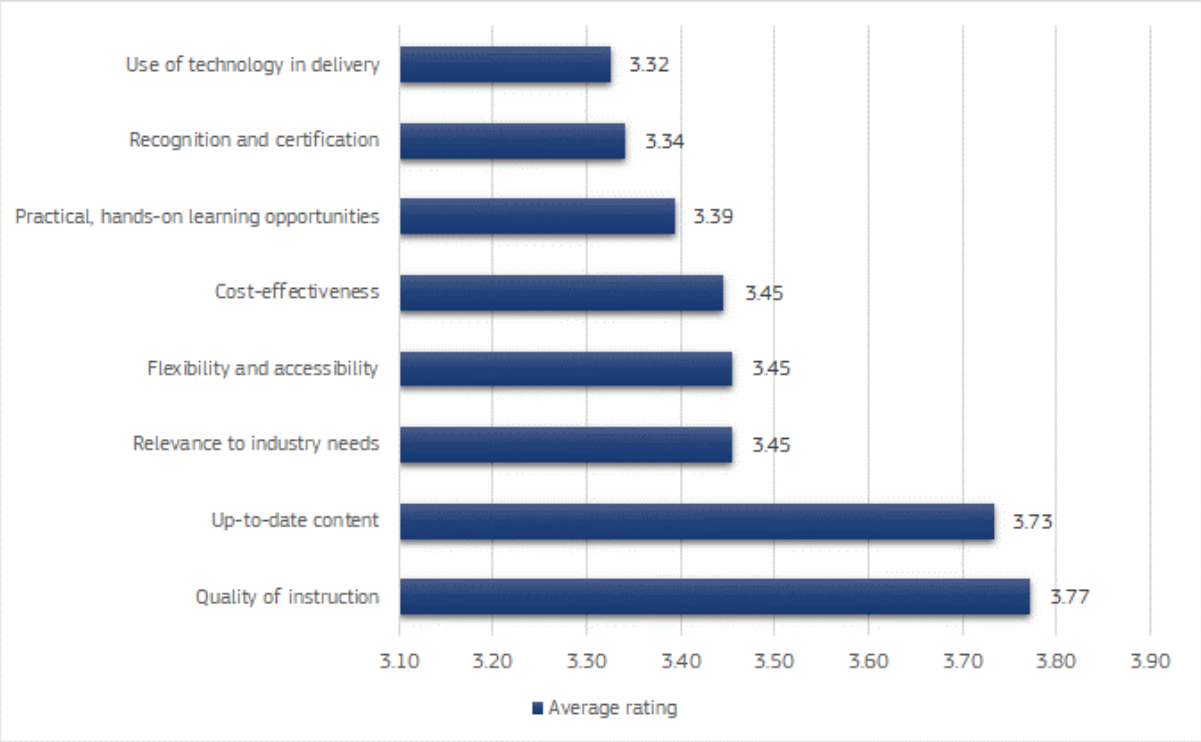


Q21. *Considering all the training programmes you are familiar with, rate the following aspects:*

In this question, participants were asked to evaluate different aspects of training programmes on a scale from 1, poor, to 5, excellent.

Quality of instruction and **up-to-date content** emerged as the highest-rated aspects, both receiving an average score above 3.5, reflecting positively on the relevance and expertise of the training programmes. On the other hand, the **use of technology in delivery** received the lowest ratings, aligning with findings from Q22 and Q23, which identified this as a key area requiring improvement. Additionally, practical, **hands-on learning opportunities** also scored low, highlighting a potential gap between theoretical instruction and real-world application, complementing the results in Q20a.

Figure 11 Average rating for training programmes' aspects



Q22. *What aspects of existing blue skills education and training programmes could be improved?*

For this question, participants could choose more than one answer. Three factors, **collaboration between industry and education providers**, **practical, hands-on learning opportunities**, and **curriculum relevance to industry needs**, which were indicated as area of improvement by, respectively, 70%, 50% and 49% of the respondents, collectively highlight a possible gap between the real-world requirements of the blue economy and the current state of training programmes, already spotted in the previous questions.

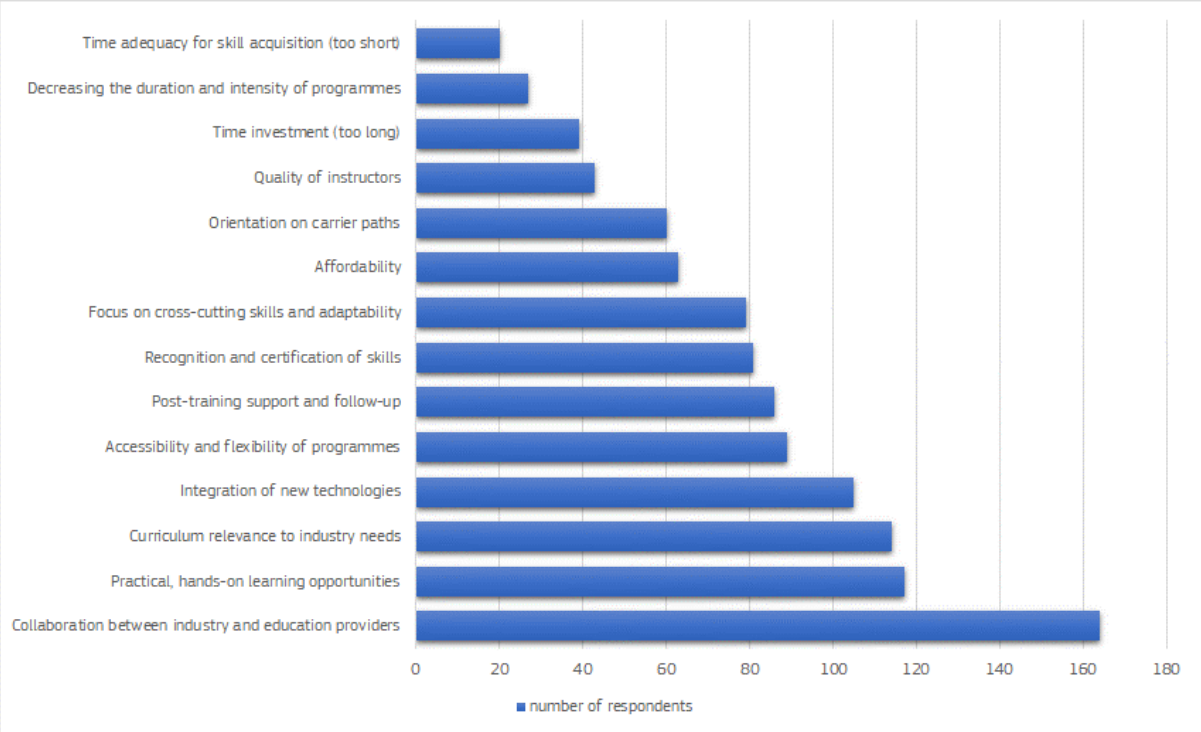
Integration of new technologies placed fourth as a possible area of improvement, as it was indicated by 45% of the participants. It reflects the growing importance of technological advancements in the blue economy and a perceived lag in integrating these technologies into training.

The last three factors shown in the chart below—**time investment (too long)**, **decreasing the duration and intensity of the programmes** and **time adequacy for skill acquisition**—received the lowest response rates, indicating that they are not perceived as significant gaps in current training programmes. Similarly, **quality of instructors**, with only 18% of respondents identifying it as an area for improvement, reinforces this positive assessment of the current training structure.

When examining sectors, **Blue Biotechnology** presents interesting results. It has the highest rate of respondents highlighting the need for integration of new technologies and also reports a substantial number of responses pointing to the importance of **collaboration**

between industry and education providers and **practical, hands-on learning opportunities**. This may be related to the highly technical nature of the sector, which needs to keep pace with evolving technologies.

Figure 12 Areas of improvement for training programmes



Q23. *What innovative training methods or approaches do you think could enhance blue skills development?*

The survey reveals a strong emphasis on innovative, immersive, and practical training methods to enhance blue skills development. Respondents frequently suggested the use of **advanced technologies such as Virtual Reality (VR), Augmented Reality (AR), and simulation-based learning**, which provide realistic and interactive scenarios to bridge the gap between theoretical knowledge and practical application. This approach aligns perfectly with the survey's broader call for more **hands-on fieldwork in blue skills curricula**, which topped the list of suggestions in this section. The benefits of VR and AR in this context include the ability to simulate realistic scenarios without the associated risks, accelerate learning curves for complex systems, and provide training in exceptional or rare situations. In essence, virtual hands-on fieldwork is made possible by the proliferation of new technologies, offering an immersive experience that mirrors real-world environments, but in a controlled and safe digital space.

Gamification was also identified as a promising approach to make learning more engaging, alongside the integration of blended learning techniques, combining online theoretical instruction with in-person practical fieldwork. Many respondents emphasised the importance of aligning training with the evolving needs of the blue economy through personalised

learning paths and **stronger collaborations between academia and industry**. Initiatives such as **internships, apprenticeships, and partnerships** with blue sector companies were suggested to ensure students acquire industry-relevant skills. The need for **modular** and **flexible courses** that cater to diverse participants, including working professionals, was also highlighted, as well as a focus on **upskilling and reskilling** to meet emerging industry challenges. **Sustainability and digital transformation** were frequently mentioned as crucial themes to integrate into training programmes, preparing learners to address the sector's environmental and technological challenges. Other innovative approaches included **peer-to-peer learning, community-based training,** and **mentorship programmes** to foster knowledge exchange and real-world insights. Furthermore, **micro-learning modules** and the use of mobile-friendly platforms were proposed to make training accessible to broader audiences, particularly in remote or underserved regions.

Table 8 Training methods, frequency mentioned

Training method	Frequency mentioned
Hands-on fieldwork	18
Virtual Reality (VR) & Augmented Reality (AR)	15
Blended learning	14
Simulation-based training	12
E-Learning platforms	11
Gamification	10
Collaboration between academia and industry	9
Microlearning	8
AI & machine learning tools for training	8
Sustainability-focused training	7

Q24. *Are you aware of any best practices or particularly successful blue skills training programmes?*

Q24 was on open-text question to which 41 participants replied. Among these, the contractor identified 20 distinct training programmes that respondents considered successful, as shown in Table 9 below. Some of these programmes were already scraped and included in our database, with the list provided here for reference.

Table 9 Best practices and or successful training programmes

Training programme	Focus or description	Key elements	EU funding
IMO's MTCC Programme	Capacity-building for energy efficiency and emissions reduction	Provides targeted training and technical assistance to support sustainable shipping globally.	✓

Training programme	Focus or description	Key elements	EU funding
WE are Blue Schools Project	Blue economy principles education in secondary schools	Encourages young people to develop sustainability skills, deepening their knowledge in ocean literacy.	✓
World Maritime University Courses	Maritime policy, ocean governance, port management	Provides leadership skills for driving sustainable practices in maritime sectors.	X
Next BlueGeneration Project	Raising awareness about blue careers opportunities	Inspiring young people to explore opportunities in blue careers.	✓
OceanWise Programme by UNESCO	Ocean data and resource management skills	Empowers coastal nations with tools to manage marine resources effectively.	X
Australian Maritime College Offshore Safety Training	Offshore survival skills and emergency response	Realistic simulations ensure high levels of preparedness.	X
ProBlue Initiative by the World Bank	Sustainable marine resource use through skills development	Tailored training modules for developing nations, fostering economic resilience.	X
BOUTCAR	Developing an educational path in the fisheries and aquaculture sector	Shapes skilled people needed to support EU blue economy initiatives.	✓
Turning Blue Project	Integrating young offenders into the blue economy	Engages employers and leads to job retention in the blue economy.	✓
European Ocean Literacy Coalition (EU4Ocean)	Promoting ocean literacy and sustainable ocean management	Improves ocean literacy and promotes responsible choices for ocean protection.	✓
FLORES-Pact for Skills	Training workers for offshore renewable energy	Promotes the skilling process for the new jobs in the sector.	✓
GWO Training for Wind Energy Workers	Specific offshore wind energy training	Includes a combination of theoretical knowledge and hands-on experience.	X
Blue Bio Value Programme	Supporting sustainable marine biotechnology solutions	Aims to develop an economic model based on marine biotechnology.	X
Port Virtual Lab	Simulated logistics operations	Allows participants to simulate and solve complex logistics challenges.	X

Training programme	Focus or description	Key elements	EU funding
Catching the Potential (EU)	Sustainable fisheries training	A European standard for sustainable fisheries training, with country-specific adaptations.	✓
Mares Programme (France)	Marine environmental training	Focuses on sustainable maritime practices.	X
NIFI Programmes (Canada)	Fisheries and seafood training	Includes practical skills development for the fisheries sector.	X
Fit4Blue Project	Fishing tourism and sustainable practices	Includes training on sustainable practices in fishing tourism.	✓
WINBLUE Project	Gender equality in the blue economy	Empowers women in the blue economy, focusing on gender equality.	✓
EU-CONEXUS	MSc in marine biotechnology	A joint MSc programme across universities, focusing on marine biotechnology education.	✓

5. Barriers and solutions

In this section, participants were asked to identify the barriers and solutions they see to skills development in the blue economy from a set list of possible answers, plus the option to tick "Other" and thus suggest more barriers and solutions not included in the list. Of note, out of 233 respondents, only 7 ticked "Other" barriers, whereas 3 suggested "Other" solutions. This implies that the list provided by the survey questionnaire resonated well with blue economy stakeholders who participated to this survey.

General findings

The survey responses revealed several key barriers to skills development in the blue economy. A primary challenge was the difficulty in attracting and retaining talent, which was closely linked to a lack of awareness about career opportunities and unclear career pathways. These workforce-related issues were seen as central to the skills gap in the sector. Limited funding for training and upskilling was another critical barrier, compounded by time constraints that made it difficult for individuals to pursue further education or training. For potential solutions to overcoming barriers to skills development in the blue economy, respondents emphasised the importance of strengthening collaborations between industry and educational institutions. Increased funding for training and upskilling was also identified as a crucial need, directly addressing the financial constraints that limit access to education in the sector.

Q25. In your opinion, what are the barriers to skills development in the blue economy?

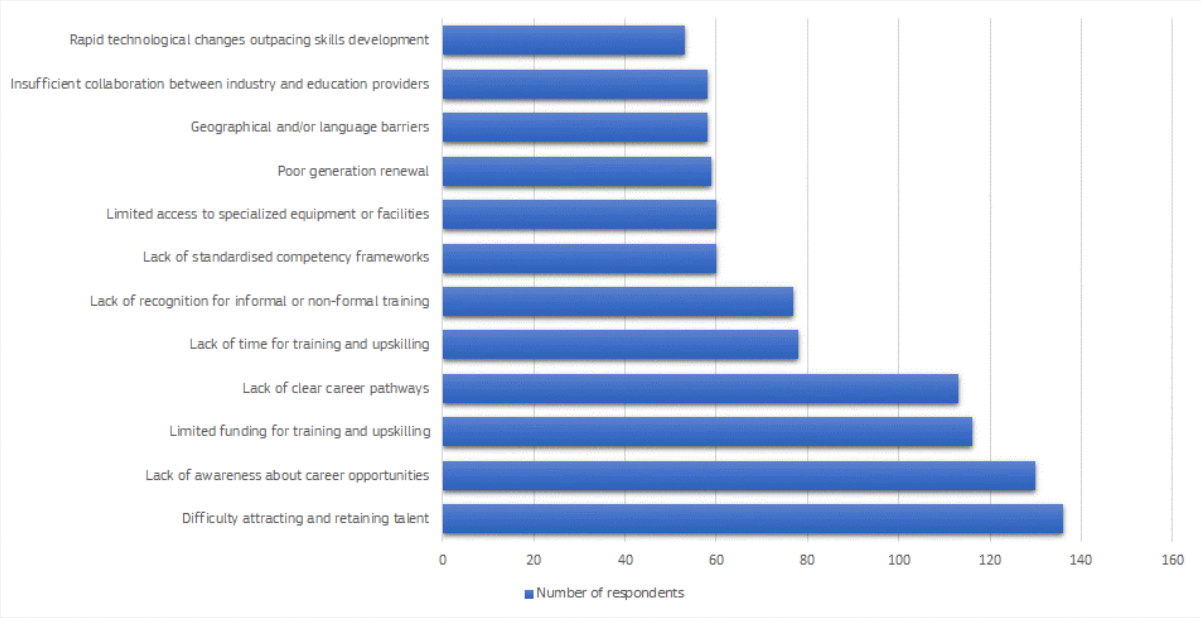
For this question, participants could choose more than one answer and, interestingly enough, all the possible answers revealed themselves to be particularly relevant, considering that even the least and second least chosen, namely **rapid technological changes outpacing skills development** and **insufficient collaboration between industry and education providers** were noted by 23% and 25% of participants, respectively, slightly less than one fourth of respondents.

58% of respondents identified the **difficulty** in **attracting and retaining talent** as a barrier. This concern was widespread across most sectors, with the main exceptions being **Marine Renewable Energy** and **Offshore Wind Energy**, and it is closely tied to two other barriers: a **lack of awareness about career opportunities**, highlighted by 56% of participants and most felt in Offshore Wind Energy and Shipbuilding and Repair, and the **lack of clear career pathways,** by 48%, most felt in Maritime Transport, and Blue Biotechnology. Notably, these barriers are all workforce-related, indicating that human resource issues are at the core of the skills gap in the blue economy.

Limited funding for training and upskilling was another critical barrier, with 50% of respondents highlighting it as an issue. This aligns with responses from Q26, where increased funding was identified as a key solution to overcoming blue skills challenges. Finally, 33% of participant flagged the **lack of time for training and upskilling** as a barrier. This barrier could be addressed by incorporating **microlearning**, a strategy suggested in Q23, which

consists in the delivery of content in short, focused segments, making it easier for workers to acquire knowledge quickly and efficiently, without requiring large time commitments.

Figure 13 Barriers to skills development



Q26. *In your opinion, what are the potential solutions to overcoming barriers to skills development in the blue economy?*

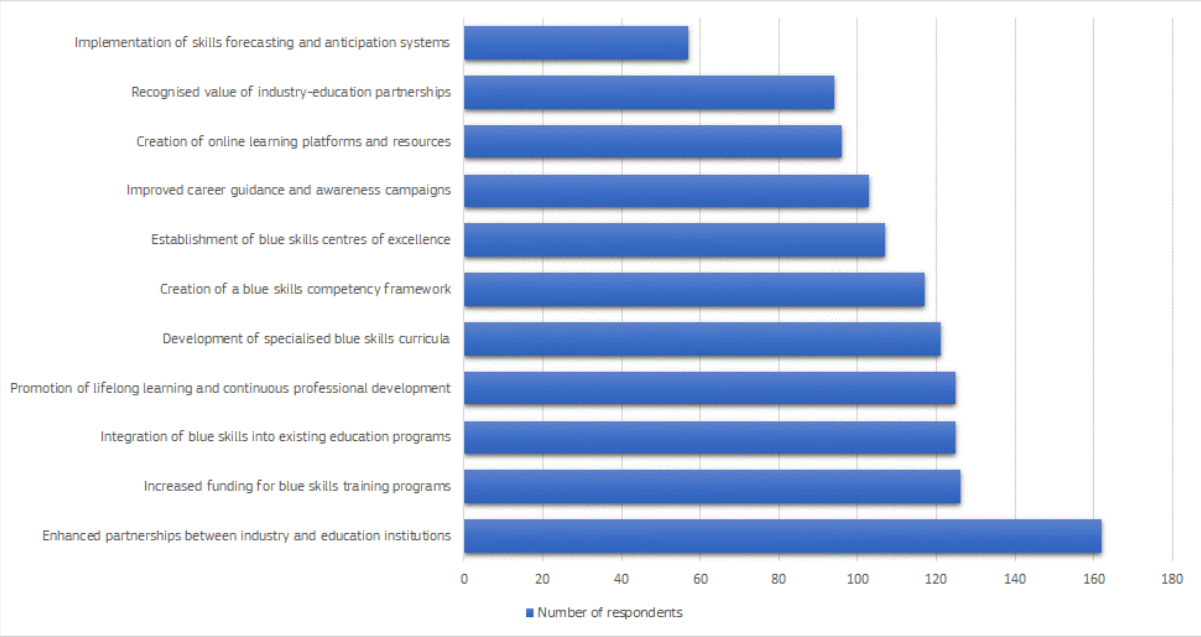
For this question, participants could select multiple answers, and the findings suggest a strong consensus on the potential solutions to overcome barriers in blue skills development.

The most prominent solution, supported by 70% of respondents, was the **enhancement of partnerships between industry and educational institutions**. This solution aligns with the findings of §5 of the report, yet an interesting mismatch arises when comparing it with the responses to Q25. While **insufficient collaboration between industry and education providers** was not flagged as a primary barrier by the majority of respondents in Q25, 70% viewed enhanced collaboration as a viable solution in Q26. This discrepancy could be interpreted in view of the subtle distinction between "insufficient" and "enhanced". It suggests that while the existing level of collaboration may be seen as sufficient, there is widespread support for further strengthening these partnerships to help address barriers such as **difficulty** in **attracting and retaining talent**, and **raising awareness about career opportunities in the sector.** The second most popular solution, with 54% of respondents, was **increased funding for blue skills training programmes**, which directly correlates with the barrier identified in Q25: **limited funding for training and upskilling**.

Interestingly, the **integration of blue skills into existing educational programmes** was favoured over the **development of specialised blue skills curricula**, with 54% and 52% of respondents, respectively, choosing these options. While the margin is small, it suggests a preference for integrating blue skills into broader educational frameworks rather than

developing entirely specialised curricula. This reflects the interdisciplinary nature of blue skills, which often require a combination of expertise from various fields. The ideal solution might therefore involve blending specialised training with broader educational content, providing a well-rounded skill set that meets the diverse needs of the blue economy.

Figure 14



6. Definition of “blue skills”

One of the primary objectives of this work is to arrive at a coherent, comprehensive, and shared definition of blue skills. In this survey, participants were asked to provide feedback on the proposed definition, contributing their perspectives as stakeholders in the blue economy to help refine and validate it, ensuring it aligns with the sectors’ realities and needs.

Here is the proposed definition:

“Blue skills” refers to the comprehensive set of knowledge, values, attitudes, competencies, and qualifications that workers should acquire to live in, develop and support the sustainable development and growth of the blue economy.

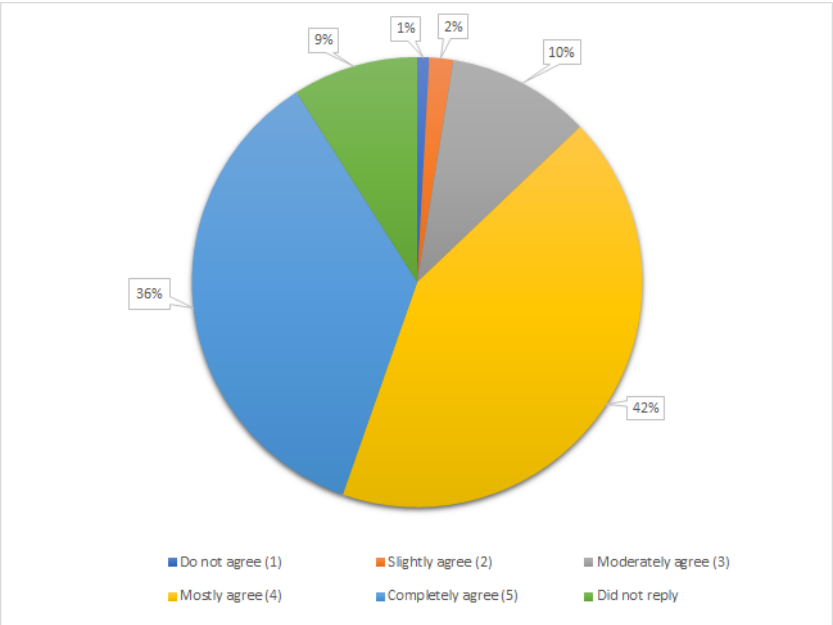
These skills encompass a wide range of disciplines, and cover technical expertise specific to each sector, as well as cross-cutting competencies such as ocean literacy, environmental sustainability awareness, proficiency in digital technologies, and the ability to innovate. The development of blue skills involves various levels of education and training, from vocational programmes to higher education and research. The concept of lifelong learning is integral to blue skills development, recognising that the marine sector's dynamic nature necessitates continuous skill updating and knowledge acquisition throughout one's career.

Q27. To what extent do you agree with the definition above?

In this question, participants were asked to evaluate their agreement with the proposed definition of blue skills on a scale from 1, do not agree, to 5, completely agree, and to leave the question blank if they did not have an opinion.

As can be seen in the chart below, the majority of respondents mostly or completely agreed – 78% of the total – with the proposed definition, while only 3% disagreed or only slightly agreed.

Figure 15



Q27a. Are there any concepts missing in the definition? Which ones? We encourage you to define blue skills in your own words.

The participants who disagreed or only slight agreed with the definition of “blue skills”, 3% of the total, emphasised the need for better clarity. Here below is a summary of their comments:

- The definition should provide greater clarity and detail, particularly on the technical skills required in various maritime activities (e.g., fisheries, offshore wind, tourism).
- Terms like “ocean literacy" were considered too complex, with suggestions to simplify or replace them.
- Specialised skills and adaptability, especially in evolving fields like offshore wind, were seen as essential but underemphasised.
- The need to align the definition with a sustainable blue economy, where sustainability is a core principle.
- The term “blue skills” was viewed as too narrow by some, with suggestions to use “blue competences” or explicitly reference the maritime sector.

This issue highlights a broader challenge inherent to this project. Both the survey results and the rest of the work reveal that the term “skills” is traditionally used, but in practice, it often refers to what are more accurately defined as “competences.” The distinction between the two terms is important: skills generally refer to the ability to perform specific tasks or actions, often tied to practical or technical know-how. In contrast, competences encompass a broader range of capabilities, combining knowledge, skills, and attitudes needed to perform effectively in a given context or role. Competences also imply a level of autonomy and responsibility, making them more aligned with the comprehensive expertise required in many roles within the blue economy. Despite this distinction, the term “blue skills” has been widely adopted and consistently used in discussions and policies. To maintain clarity and consistency with existing literature and terminology, we will continue to use “blue skills” in this report, and in this project. However, it should be understood that this term is used interchangeably to refer to both skills and competences, reflecting the comprehensive capabilities needed across blue economy sectors. The definition should avoid relying on prior knowledge of terms like "blue" and “blue economy” and instead clearly define its relationship to the maritime context.

The rest of the respondents, 78% of the total who moderately to fully agreed with the definition, expressed opinions that were not vastly different from those who rated it lower, as some themes were shared. Common concerns included the need for a more nuanced and comprehensive approach to sustainability, encompassing environmental, social, and economic dimensions. Several respondents emphasised that the definition of blue skills should reflect a sustainable, inclusive, and interdisciplinary approach, incorporating expertise from marine sciences, technology, economics, and social sciences. There is a call for better recognition of community engagement, stakeholder participation, and local knowledge, highlighting the importance of integrating these elements into both the development and implementation of blue economy practices. Many also noted the importance of adapting to

emerging challenges such as climate change, technological advancements, and the need for continuous, lifelong learning. It was also suggested that the definition should stress the importance of specialised technical skills, adaptability, and the capacity for innovation, especially in fields like offshore wind energy and aquaculture. Additionally, several responses pointed out the importance of defining "blue economy" more clearly, as its scope is often unclear or misunderstood, which makes the definition of blue skills more challenging. Furthermore, there were suggestions to add a reference to the European Qualifications Framework (EQF) to ensure portability and recognition of qualifications across the EU. Lastly, some respondents argued that the focus should be on practical skills and work experience, which are seen as essential for successfully navigating the blue economy, with an emphasis on real-world applications and adaptability in dynamic environments.

7. Key findings

The survey results highlight sector-specific trends in skill importance, reflecting the evolving priorities and demands of the blue economy over short- and long-term horizons:

Sector-specific skills

Theoretical knowledge is increasingly valued in Aquaculture, reflecting a shift to high-tech operations, while Blue Biotechnology maintains its importance. In contrast, Infrastructure and Offshore Wind Energy are moving towards greater emphasis on practical skills. Practical knowledge is becoming more critical in Offshore Wind Energy and Shipbuilding, with Port Activities consistently relying on it. Regulatory compliance sees a sharp rise in importance for Blue Biotechnology, while Offshore Wind Energy and Port Activities indicate a decline, suggesting changing operational priorities. Risk prevention remains highly valued in Coastal Tourism and Shipbuilding, but declines significantly in Infrastructure, with Blue Biotechnology gradually increasing its focus. These trends underscore evolving sector-specific demands driven by innovation, operational needs, and safety.

General digital skills

General digital skills grow significantly in Shipbuilding (+23%) and Offshore Wind Energy (+10%), while Coastal Tourism and Maritime Transport remain consistently high. Advanced digital skills see strong growth in Coastal Tourism (+28%) and Marine Renewable Energy (+25%), with Port Activities maintaining high importance. Information and data literacy rises in Shipbuilding (+22%) and Coastal Tourism (+15%), while Infrastructure and Port Activities improve slowly. Digital communication grows modestly in Fisheries and Shipbuilding but declines in Offshore Wind Energy (-15%). Cybersecurity importance grows in Marine Renewable Energy (+19%) and Maritime Transport, while Port Activities remain the highest priority. These shifts highlight the increasing importance of digital skills for innovation and sustainability across sectors.

Green skills

Environmental sustainability knowledge sees notable growth in Infrastructure and Robotics (+26%) and Coastal Tourism (+15%), while Blue Biotechnology maintains high importance. Port Activities and Offshore Wind Energy experience slight declines in the long term, indicating shifting priorities. Environmental management and conservation skills gain traction in Infrastructure (+33%) and Shipbuilding (+22%), with Port Activities achieving a perfect score of 5.0. Offshore Wind Energy shows a slight decrease, contrasting with steady growth in Marine Renewable Energy. Circular economy principles are increasingly valued in Infrastructure, Shipbuilding, and Coastal Tourism, all showing significant growth (+24%).

Soft and cross-cutting skills

Project management grows in Infrastructure (+16%) and Aquaculture (+13%), with Blue Biotechnology emphasising complexity. Communication and teamwork are prioritised in Aquaculture (4.90) and Coastal Tourism (+8%). Innovation and problem-solving excel in Aquaculture (+12%) and Port Activities (5.0), though Infrastructure declines slightly. Adaptability and resilience surge in Infrastructure (+23%) and Fisheries (+15%), while cross-cultural competence sees sharp increases in Infrastructure (+28%) and Fisheries (+20%). Risk management grows strongly in Infrastructure (+29%) and Fisheries (+20%), while ocean literacy rises in Fisheries (+17%) and Maritime Transport (+16%), with Coastal Tourism maintaining high ratings.

Business and managerial skills

Financial management rises significantly in Infrastructure (+33%), Blue Biotechnology (+19%), and Coastal Tourism (+15%), with Port Activities remaining the highest priority (4.25). Marine Renewable Energy and Maritime Defence see declines. Customer service grows strongly in Infrastructure (+20%) and Blue Biotechnology (+16%), while Port Activities and Coastal Tourism stay steady. Offshore Wind Energy drops sharply (-9%), reflecting reduced focus on customer roles. Leadership and strategic thinking surges in Infrastructure (+23%), with Port Activities (4.75) and Aquaculture (4.50) leading. Coastal Tourism and Fisheries (+16%) also show strong gains, reflecting increased emphasis on strategic leadership across sectors.

Skills identified by respondents

Skills identified by respondents as important for their sectors but not included in the predefined list: bioinformatics (Blue Biotechnology sector), downstream processing (Blue Biotechnology and Marine Renewable Energy sectors), decarbonisation and carbon footprint management (Maritime Transport sector), genomics and omics (Blue Biotechnology sector), GIS (Aquaculture, Fisheries and Offshore Wind Energy sectors), alternative fuels (Maritime Transport and Marine Renewable Energy sectors).

New jobs roles and occupations

The survey highlights a transformative shift in blue economy roles, driven by technology, sustainability, and industry challenges. Emerging positions like AI Specialists, Blue Carbon Advisors, and Marine Biotechnologists reflect growing demands for advanced expertise and environmental focus. Cross-sector roles, such as Energy Transition Managers and Marine Spatial Planners, underscore interdisciplinary needs, while strategic roles like Blue Economy Policy Advisors will shape sustainable maritime policies. These trends emphasize the evolving complexity and innovation required in the blue economy workforce.

Education and training programmes

The findings show 58% of respondents are aware of blue skills training programmes, with high awareness in Maritime Transport, Fisheries, and Aquaculture. Participation in blue skills training is widespread, with 63% of respondents involved in the past two years. Common formats include workshops and webinars, while apprenticeships and mentoring are underutilised despite their importance. A gap between theory and practical training was noted, with calls for more real-world applications. Respondents emphasised industry-education collaboration, hands-on opportunities, and curricula aligned with sector needs as areas for improvement. Technologies like Virtual and Augmented Reality, along with gamification and blended learning, were suggested to enhance training engagement and accessibility.

Barriers and solutions

The survey identified key barriers to blue economy skills development, including talent retention challenges due to low career awareness and unclear pathways, limited training funding, and time constraints. Solutions include stronger industry-education collaboration and increased funding to address financial barriers to training access.